

MINIPROJECT

FIRST PRESENTATION REQUIREMENTS

ML Projects

1. Literature Review

2. Data Analysis

a. Explore the Dataset

- i. Statistical Analysis: Statistical functions with its libraries and the outputs
- ii. Insights and conclusions of Exploratory Data Analysis

b. Data Preprocessing

- i. Data Cleaning: Missing values, Outliers, Balancing, Inconsistencies, Scaling, Encoding...
- ii. Feature Extraction: Methods used to identify the significant features, Apply the method(s)

c. Pre-processed Dataset

- i. Structure and characteristics of final dataset using relevant functions (head, tail, info, shape, columns, dtypes, describe etc.

3. Data visualization analysis:

Graphical functions with its libraries and visual representations

4. Explore the Algorithm / Architecture

- a. Describe the algorithm(s) and justify why that algorithms are chosen
- b. Block diagram or Flowchart of algorithm(s)
- c. Pseudocode for the algorithm(s) or Step by step explanation in plain English
- d. Describe how the algorithm(s) are used in your project (Objective of the project).

5. Project pipeline

6. GUI Design

- a. Sketch the UI proposed for deployment
- b. Interface for each screen and describe the components

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IoT Projects

1. Literature Review

- a. Review report of each reference paper
- b. Review report comprising all references
- c. Findings and Proposal

2. System Analysis

- a. Analysis of IoT Components
- b. Sensors - Pros & cons
- c. Fail safe mechanisms - What if one or multiple sensors malfunctions
- d. Circuit Diagram
- e. Sensor Placement & Configuration
- f. State Chart Diagram
- g. Project pipeline

System Design

- a. Sketch Architecture Design
 - i. How sensors, controllers, and actuators are connected
 - ii. Cloud Architecture

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DB Projects

1. Literature Review

- a. Review report of each reference paper
- b. Review report comprising all references
- c. Findings and Proposal

2. System Analysis

- a. Module Description
- b. Feasibility Analysis
- c. Technical Feasibility
- d. Economical Feasibility
- e. Operational Feasibility
- f. System Environment
- g. Software Environment
- h. Hardware Environment
- i. Actors and roles.

3. System Design

- a. Use Case Diagram
- b. Activity Diagram
- c. Sequence Diagram
- d. Class identification (classes, attributes, relationships)

- e. Class Diagram
- f. Database Design

4. GUI Design

- a. Sketch the UI proposed for deployment
- b. Interface for each screen and describe the components